

VaR and CVaR: measuring and optimizing risk



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Managerial motivation

Introduction: Managerial motivation

- ▶ In this section we see **why** optimizing an average value is not enough
- ▶ We introduce the two risk measures of the chapter: **VaR** and **CVaR**
- ▶ We list the goals: definitions, LP model, case studies, statistical limits

The average is not enough

- ▶ Many models optimize an **average value**: a simple choice, but one that can hide **rare and very large** losses
- ▶ VaR and CVaR describe the **tail** of the loss distribution and turn risk aversion into quantitative decisions
- ▶ **VaR** answers: “what is a high loss threshold?”
- ▶ **CVaR** also answers: “how much do we lose *on average* when that threshold is exceeded?”

The language of tail risk

- ▶ After the financial crises, “how much do we lose in the worst scenarios?” has become a **regulatory** question as well as a managerial one
- ▶ VaR and CVaR are the standard language with which banks, insurers and, increasingly, supply chain managers speak about tail risk
- ▶ **The didactic surprise:** CVaR, an apparently sophisticated concept, is optimized with a simple **LP**

In this chapter we will learn to

1. define and compute **VaR** and **CVaR**
2. use the **Rockafellar–Uryasev** linear formulation
3. build portfolios and logistics plans protected against the **worst scenarios**
4. recognize the **statistical limits** of tail estimates

Summary

- ▶ Optimizing the average can hide rare and large losses
- ▶ VaR: a high loss **threshold**; CVaR: the **average loss beyond** the threshold
- ▶ CVaR is optimized with an LP: this is the core of the chapter



Definitions and properties

Introduction: Definitions and properties

- ▶ We formally define the **VaR** and the **CVaR** of a random loss
- ▶ We compare their properties: why CVaR is preferable for optimization
- ▶ We build the **Rockafellar–Uryasev LP model**, which computes and optimizes the CVaR over a set of scenarios

The problem in words

- ▶ *We decide* as before: portfolio weights, quantities, booked capacity
- ▶ *The objective* changes: not to minimize the **average** loss, but rather a measure of its **tail**, that is, how much is lost in the worst scenarios
- ▶ *The constraints* remain those of the underlying problem, plus the auxiliary linear constraints that define the CVaR

Random loss and confidence level

- ▶ Let L be the **random loss** (high values = worse outcomes)
- ▶ Let $\alpha \in (0, 1)$ be a **confidence level** (typically 0.90–0.99)
- ▶ As for the demand of the newsvendor, the random variable is in upper case and its realizations in lower case (ℓ_s)
- ▶ This is the exception, to follow the literature, to the convention that reserves capital letters for sets and matrices

Definition 1: VaR and CVaR

$$\text{VaR}_\alpha(L) = \inf\{\eta \in \mathbb{R} : \mathbb{P}(L \leq \eta) \geq \alpha\},$$

$\text{CVaR}_\alpha(L)$ = *average loss in the worst tail
of mass $1 - \alpha$.*

Properties compared

Convexity	CVaR is convex in the decisions (if L is); VaR in general is not
Subadditivity	CVaR rewards diversification; VaR may violate it
Tail	VaR does not distinguish two tails with the same quantile but different severity
Optimization	CVaR admits a scenario-based LP formulation; VaR does not

Data of the scenario-based model

- ▶ Number of scenarios $k \in \mathbb{Z}_{\geq 1}$, indexed by $s \in \{1, 2, \dots, k\}$
- ▶ For each scenario, the probability $\pi_s \in \mathbb{Q}_{\geq 0}$, with $\sum_{s=1}^k \pi_s = 1$
- ▶ For each scenario, the loss $\ell_s(\mathbf{x})$, a linear function of the decision vector $\mathbf{x}$, constrained to a polyhedron X
- ▶ The confidence level $\alpha \in (0, 1)$

Decision variables

Besides the decisions $\mathbf{x}$, we introduce one free variable and k non-negative variables:

$$\begin{cases} \eta = \text{candidate loss threshold (at the optimum: a VaR)} \\ \xi_s = \text{loss in excess of } \eta \text{ in scenario } s \end{cases}$$

$$\forall s \in \{1, 2, \dots, k\}.$$

Definition 2: Rockafellar–Uryasev LP model for the CVaR

$$\min \quad \eta + \frac{1}{1 - \alpha} \sum_{s=1}^k \pi_s \xi_s$$

$$s.t. \quad \xi_s \geq \ell_s(\mathbf{x}) - \eta, \quad \forall s \in \{1, \dots, k\},$$

$$\mathbf{x} \in X,$$

$$\eta \geq 0,$$

$$\xi_s \geq 0, \quad \forall s \in \{1, \dots, k\}.$$

Description of the model (1/2)

- ▶ **Objective function:** the threshold η plus the weighted average of the tail excesses, amplified by $1/(1 - \alpha)$
- ▶ *At the optimum η^* is a VaR and the objective value is the CVaR: a single LP provides both measures*
- ▶ **Tail constraints** $\xi_s \geq \ell_s(\mathbf{x}) - \eta$: they define the excesses; only the scenarios with a loss beyond the threshold contribute (k linear constraints)

Description of the model (2/2)

- ▶ **Constraint $\mathbf{x} \in X$:** collects the constraints of the underlying decision problem (polyhedral)
- ▶ **Definition of the variables:** η is *free* (the symbol $\gtrless$), the ξ_s are non-negative

Why the model works

- ▶ The variable $\xi_s = (\ell_s - \eta)^+$ is the loss in excess of the candidate threshold η (the same positive-part trick as in the Newsvendor)
- ▶ Minimizing over η balances two forces: raising η reduces the excesses but pushes up the threshold
- ▶ The equilibrium point is exactly the α **quantile**
- ▶ If X is a polyhedron and ℓ_s is linear, the whole model is an **LP**

Summary

- ▶ VaR_α : α quantile of the loss; CVaR_α : average loss in the tail of mass $1 - \alpha$
- ▶ CVaR is convex, subadditive, sensitive to the severity of the tail and optimizable with an LP; VaR is not
- ▶ Rockafellar–Uryasev model: free threshold η , excesses $\xi_s \geq 0$, k tail constraints
- ▶ At the optimum: η^* is a VaR, the optimal value is the CVaR

Worked example by hand

Introduction: Worked example by hand

- ▶ We compute VaR and CVaR by hand on an example with **six scenarios**
- ▶ We check that the Rockafellar–Uryasev LP returns the same values
- ▶ We compare mean, VaR and CVaR: three numbers, three stories

Example: six scenarios, $\alpha = 0.80$ (1/3)

- ▶ Equally likely losses: $\{2, 4, 5, 7, 12, 20\}$, each with probability $1/6$
- ▶ **VaR**. The cumulative distribution reaches 0.80 at the fifth ordered loss:

$$\text{VaR}_{0.80} = 12$$

Example: six scenarios, $\alpha = 0.80$ (2/3)

- ▶ **CVaR with the formula.** The worst tail has mass 0.20
- ▶ The scenario with loss 20 weighs $1/6 = 0.167$, and the remaining mass 0.033 sits *on the point* 12:

$$\text{CVaR}_{0.80} = 12 + \frac{1}{0.20} \cdot \frac{1}{6} (20 - 12) = 12 + 6.67 = 18.67$$

Example: six scenarios, $\alpha = 0.80$ (3/3)

- ▶ **Check with the LP** (script): $\eta^* = 12$, $\xi = (0, 0, 0, 0, 0, 8)$, value 18.6667
- ▶ The formula correctly handles the mass “straddling” the quantile without selecting scenarios with binary variables
- ▶ **Comparison.** Average loss 8.33; VaR 12; CVaR 18.67
- ▶ The mean reassures, the VaR gives a threshold, the CVaR says how much it hurts when things go badly

Summary

- ▶ The VaR is read on the cumulative distribution: it is the loss at which it reaches α
- ▶ The CVaR averages the tail of mass $1 - \alpha$, also counting the mass that sits on the quantile
- ▶ The LP returns the same values without binary variables

Case study 1: mean-CVaR portfolio against Markowitz



Introduction: mean-CVaR portfolio

- ▶ We apply the Rockafellar–Uryasev model to a **portfolio** of securities
- ▶ We compare the portfolio that minimizes the CVaR with the **Markowitz** one, which minimizes the variance
- ▶ We will see that, for the same return and average loss, the CVaR **cuts the tail**

The case: eight securities with fat tails

- ▶ The same eight securities as in the Markowitz chapter, but $k = 220$ monthly scenarios generated with **fat tails** (Student t market factor, 4 degrees of freedom)
- ▶ It is the world in which the CVaR shows its value
- ▶ Scenario loss: $\ell_s(\mathbf{x}) = -\sum_{i=1}^n r_{is} x_i$, where r_{is} is the return of security i in scenario s
- ▶ We minimize $\text{CVaR}_{0.90}$ with expected return $\geq 8\%$ per year

The model in Gurobi

```
eta = m.addVar(lb=-GRB.INFINITY, name="eta")  # threshold: free!
xi  = m.addVars(S, name="xi")
m.addConstr(x.sum() == 1)
m.addConstr(gp.quicksum(mu[i]*x[i] for i in range(n))
            >= r_min)
m.addConstrs((xi[s] >=
              -gp.quicksum(R[s, i]*x[i] for i in range(n))
              - eta for s in range(S)))
m.setObjective(eta + gp.quicksum(xi[s] for s in range(S))
              / (S * (1 - alpha)), GRB.MINIMIZE)
```

Results: risk measures

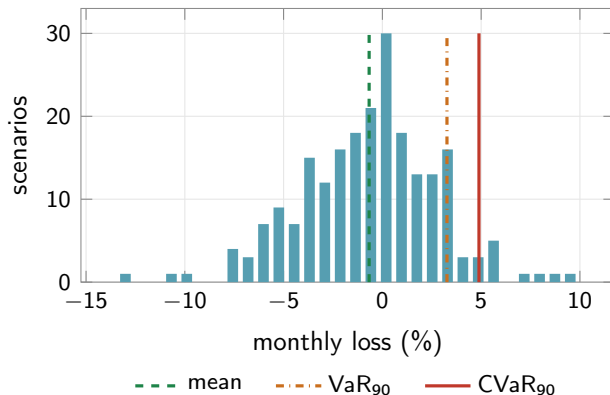
Monthly losses of the two portfolios:

	mean loss	VaR_{90}	CVaR_{90}
mean-CVaR	-0.0067	0.0327	0.0489
Markowitz	-0.0067	0.0329	0.0531

Results: compositions (%)

	ENE	FIN	TEC	SAN	UTL	MAT
mean-CVaR	9.9	1.9	10.5	8.6	46.5	22.5
Markowitz	18.4	6.4	6.5	13.6	40.4	14.6

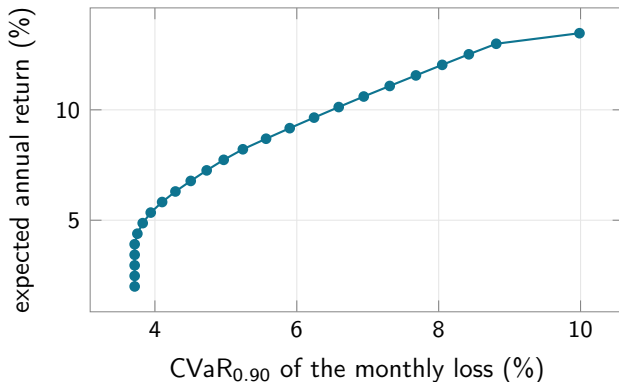
Loss distribution (mean-CVaR)



How to read the histogram

- ▶ Histogram of the monthly losses of the mean-CVaR portfolio over the 220 scenarios
- ▶ Vertical lines: **mean**, **VaR₉₀** and **CVaR₉₀**
- ▶ The CVaR is the **average of the tail** to the right of the VaR

Return–CVaR frontier



Obtained by varying the minimum required return.

CVaR cuts the tail

- ▶ For the same required return and average loss, the mean-CVaR portfolio cuts the tail: CVaR_{90} of 4.89% against 5.31% for Markowitz (-8%)
- ▶ It reduces the high-beta securities (ENE from 18% to 10%) in favour of UTL and, a surprise, of a share of TEC
- ▶ TEC, although mediocre on average, moves against the trend in the worst scenarios of this sample
- ▶ Variance penalizes symmetrically above and below the mean; the CVaR looks **only where it hurts**

Summary

- ▶ With fat tails, the mean-CVaR portfolio has a lower CVaR_{90} than Markowitz (-8%) for the same return and average loss
- ▶ The composition changes: fewer high-beta securities, more securities that protect in the worst scenarios
- ▶ Varying the minimum return traces the return–CVaR frontier

Case study 2: two-stage supply chain

Introduction: two-stage supply chain

- ▶ We use the CVaR in a **two-stage logistics** problem: first capacity is booked, then the recourse is decided
- ▶ A parameter λ balances average cost and tail risk
- ▶ We will measure the **cost of resilience**: how much average cost is needed to reduce the CVaR

The case: two suppliers

- ▶ Before observing the scenario we *book* capacity from the suppliers; then, having seen the scenario, we decide the **recourse**
- ▶ **F1** is cheap but fragile: in 12% of the scenarios its availability collapses to 30%
- ▶ **F2** is expensive but reliable
- ▶ Shortage penalty: 40 euro/unit; $k = 400$ scenarios

Data of the model

- ▶ Two suppliers $a \in \{1, 2\}$ (1 cheap but fragile, 2 expensive but reliable)
- ▶ Booking costs $c_a^{\text{pre}} \in \mathbb{Q}_{>0}$ and purchase costs $c_a^{\text{uso}} \in \mathbb{Q}_{>0}$
- ▶ k scenarios $s \in \{1, 2, \dots, k\}$ with demand $d_s \in \mathbb{Q}_{\geq 0}$
- ▶ Availability $\delta_{as} \in (0, 1]$: the datum that models the failure
- ▶ Shortage penalty $g \in \mathbb{Q}_{>0}$
- ▶ Risk weight $\lambda \in [0, 1]$ and level $\alpha \in (0, 1)$

Decision variables

Non-negative variables:

- ▶ x_a = capacity booked from supplier a : **first-stage**, the same in all scenarios
- ▶ f_{as} = purchase from supplier a in scenario s
- ▶ u_s = unmet demand in scenario s

The f_{as} and the u_s are **recourse** variables.

Two-stage model with CVaR (1/2)

Definition 3: Two-stage model with CVaR: objective

$$\min (1 - \lambda) \left[\sum_{a=1}^2 c_a^{\text{pre}} x_a + \sum_{s=1}^k \pi_s \phi_s \right] + \lambda \text{CVaR}_\alpha$$

where

$$\phi_s = \sum_{a=1}^2 c_a^{\text{uso}} f_{as} + g u_s$$

is the recourse cost of scenario s and CVaR_α is the CVaR of the total cost, linearized with the Rockafellar–Uryasev variables η, ξ_s .

Two-stage model with CVaR (2/2)

Definition 4: Two-stage model with CVaR: constraints

$$f_{as} \leq \delta_{as} x_a, \quad \forall a \in \{1, 2\}, \forall s,$$

$$f_{1s} + f_{2s} + u_s \geq d_s, \quad \forall s,$$

$$x_a, f_{as}, u_s \geq 0, \quad \forall a \in \{1, 2\}, \forall s,$$

with $s \in \{1, 2, \dots, k\}$.

Description of the model (1/2)

- ▶ **Objective function:** combines, with weight λ , the average cost (bookings plus the expected value of purchases and penalties) and the CVaR of the total cost
- ▶ **Capacity constraints** $f_{as} \leq \delta_{as}x_a$: in scenario s supplier a can be used only up to the booked capacity x_a , reduced by the availability δ_{as} ($2k$ linear constraints)

Description of the model (2/2)

- ▶ **Demand constraints:** demand is covered by the flows or, failing that, counted as a shortage u_s , which in the objective pays the penalty (k linear constraints)
- ▶ The model is never infeasible: it “buys” infeasibility at the price of the penalty
- ▶ **Last constraints:** they define the variables of the model

Results as λ varies

λ	suppl. 1	suppl. 2	mean cost	CVaR ₉₀	service
0.0	161.5	134.2	1,724	2,902	98.7%
0.5	92.7	212.7	1,803	2,217	99.9%
1.0	68.5	236.9	1,906	2,139	99.0%

Columns suppl. 1 and suppl. 2 report the booked capacity.

The cost of resilience

- ▶ The risk-neutral decision maker ($\lambda = 0$) buys from the cheap supplier and accepts that in 12% of the scenarios the shortage makes costs explode (CVaR 2,902)
- ▶ As λ grows, capacity migrates towards the reliable supplier
- ▶ +79 euro of average cost buy –685 euro of CVaR
- ▶ This is the **cost of resilience**: the number needed in a discussion about reshoring and dual sourcing

Summary

- ▶ Two stages: capacity booked before the scenario, purchases and shortage after it
- ▶ The objective combines average cost and CVaR with weight λ ; the shortage makes the model always feasible
- ▶ As λ grows, capacity moves to the reliable supplier: the CVaR drops much more than the average cost rises

Sensitivity analysis

Introduction: Sensitivity analysis

- ▶ We see which parameters to vary and with which expected effect
- ▶ We warn about the **statistical limits** of tail estimates

Parameters to vary

-
- | | |
|-----------|---|
| α | higher $\Rightarrow$ a more extreme tail estimated from fewer scenarios: more scenarios are needed for stable estimates |
| λ | controls the trade-off between average performance and protection |
| π_s | reweighted scenarios or stress tests (adding extreme scenarios) |
| $ S $ | statistical stability: repeat with different samples, as in the newsvendor chapter |
-

Beware of tail estimates

Observation 1: Tails estimated from few scenarios

With $\alpha = 0.99$ and $k = 220$ scenarios the tail contains only 2–3 scenarios: the estimated CVaR is almost noise.

Rule of thumb: at least a few dozen scenarios are needed beyond the quantile.

With discrete distributions the VaR may not be unique (η^ is a VaR): always compare η^* with the empirical quantile.*

Summary

- ▶ α , λ , π_s and the number of scenarios are the levers of the analysis
- ▶ Extreme tails with few scenarios give noisy estimates: dozens of scenarios beyond the quantile are needed
- ▶ η^* is a VaR: compare it with the empirical quantile

Exercises

Introduction: Exercises

- ▶ Five exercises to consolidate the concepts of the chapter
- ▶ They range from computation by hand to modifying the models of the case studies, to be solved with the lab script

Exercise: check

- ▶ Recompute by hand the VaR and CVaR of the 6-scenario example with $\alpha = 0.90$
- ▶ Check with the LP
- ▶ Careful: the tail has mass 0.10 and the scenario with loss 20 weighs $0.167 > 0.10$

Exercise: subadditivity

- ▶ Build a scenario-based example with two activities in which

$$\text{VaR}(L_1 + L_2) > \text{VaR}(L_1) + \text{VaR}(L_2)$$

- ▶ Check that the CVaR does not violate subadditivity

Exercise: frontier in α

- ▶ For the portfolio, plot the optimal CVaR for

$$\alpha \in \{0.80; 0.90; 0.95; 0.99\}$$

at equal return

- ▶ How does the composition change?
- ▶ When do the numerical instabilities begin?

Exercise: CVaR-constrained

- ▶ Reformulate the portfolio as

$$\max \sum_{i=1}^n \mu_i x_i \quad \text{subject to} \quad \text{CVaR}_{0.90} \leq \bar{c}$$

- ▶ Check that by varying $\bar{c}$ one traces the same frontier
- ▶ Read the dual of the constraint

Exercise: stress test

- ▶ Add to the supply chain an extreme scenario (demand +50% and F1 at 10%) with probability 1%
- ▶ How much does the optimal booking change for $\lambda = 0.5$?
- ▶ And the CVaR?

Final summary (1/2)

Risk measures:

- ▶ VaR: loss threshold (α quantile); CVaR: average loss in the tail of mass $1 - \alpha$
- ▶ CVaR is convex, subadditive and optimized with an LP

Rockafellar–Uryasev model:

- ▶ Free threshold η and excesses $\xi_s \geq \ell_s(\mathbf{x}) - \eta$
- ▶ At the optimum η^* is a VaR and the optimal value is the CVaR

Final summary (2/2)

Case studies:

- ▶ Portfolio: the mean-CVaR portfolio cuts the tail compared with Markowitz
- ▶ Supply chain: λ measures the cost of resilience

Limits:

- ▶ Extreme tails with few scenarios give noisy estimates
- ▶ η^* is a VaR: compare it with the empirical quantile